

1 • Shared evidence: delivery routes

The nutrient delivery investigation

Name _____ Date _____

Use these science cards for paper tokens or the interactive model. No measurements from pupils are needed.

A • Digestion

Some large food molecules are broken into small molecules. Glucose is a small soluble sugar.

C • Water

Water is absorbed in both small and large intestine. It does not need to be digested into a smaller substance.

E • Unabsorbed material

Material that is not absorbed continues through the large intestine. Faeces are stored in the rectum and leave through the anus.

B • Absorption

Most nutrients are absorbed in the small intestine. Glucose crosses its wall into blood; most dietary fat initially takes a lymph route.

D • Transport

Blood plasma is mostly water and carries many dissolved substances. Circulation moves them towards tissues.

F • Kidneys

Kidneys filter blood and help balance water and dissolved substances. They make urine, which is carried to and stored in the bladder.

W1. Choose the process for large starch, glucose at gut wall, glucose in blood and water in gut.

W2. Repair “drink → stomach → bladder”.

W3. Explain the route of unabsorbed material and how urine differs.

2 • Supported route: follow the substance

Use page 1. Word bank: absorption / digestion / transport / blood / kidneys.

S1. Which process breaks some large food molecules down?

- A. Digestion
- B. Absorption

S2. Complete: Glucose crosses the gut wall by _____. It is carried in _____ towards tissues.

S3. Does water have to be digested before it is absorbed?

- A. Yes
- B. No

S4. Draw a route for water: intestine → _____ → body tissues. Label the crossing step.

S5. Complete: _____ make urine from blood. Unabsorbed material continues through the _____. Word bank: gut / kidneys.

S6. Explain one difference between digestion and transport.

3 • Standard route: two delivery records

Use precise substance and process names.

Process	Meaning in this lesson
Digestion	
Absorption	
Transport	

N1. Complete the table, then give one example of absorption.

N2. Trace glucose from the small intestine across its wall and towards a body tissue. Label each process.

N3. Trace absorbed water towards tissues. Explain why digestion is not needed for water.

N4. Compare the faeces route with the urine route. Include the kidneys.

4 • Stretch route: audit the delivery animation

An accurate model states what it shows and what it simplifies.

T1. Repair “Whole food travels through blood to muscles”. Use digestion, absorption and transport accurately.

T2. Explain a water journey and why “all swallowed water goes directly to the bladder” is wrong.

T3. A diagram sends every dietary nutrient straight into blood. Use card B to name a limitation and improve its label.

T4. Explain why a gut-wall diagram is not a literal kitchen sieve. State what its arrows represent.

5 • Exit and reflection

A labelled route is welcome.

E1. What is absorption?

E2. What transports many dissolved nutrients around the body?

E3. Which organs make urine by filtering blood?

E4. Does water need digestion before absorption? Explain.

Reflection. Which route would you like to magnify and why?
